

Lie & Emotion Detector

Real-time Deception Detection via Facial Expression Analysis

Mobile Computing and Its Applications (4190.406B) — Spring 2026

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1 Goal & Usage Scenario

The goal of this project is to build a real-time mobile app that detects whether a user’s displayed emotion is genuine, and if not, infers the underlying true emotion using camera input alone.

Final output:

- Is this emotion **genuine or deceptive**? (True / False)
- If deceptive, what is the **true emotion**? (e.g., angry, sad, ...)

Usage scenarios: interview practice, negotiation, social skills training, or simply checking whether your friend is really fine.

2 Related Works

Micro-expression recognition. CASME II [1] and SAMM [2] are widely used benchmarks for spontaneous micro-expression detection, where subtle involuntary expressions reveal concealed emotions.

Deception detection. The MU3D dataset [3] contains videos of genuine and deceptive facial expressions across multiple emotion categories, making it well suited as a supervised signal for deception classification.

Semi-supervised learning. FixMatch [4] achieves strong performance with few labeled examples by assigning pseudo-labels to unlabeled data based on a confidence threshold. SimCLR [5] learns transferable visual representations via contrastive learning without labels.

Lightweight on-device inference. MediaPipe Face Mesh [6] extracts 468 3-D facial landmarks in real time on mobile devices, enabling lightweight coordinate-based DNN inputs instead of raw image inputs.

3 Key Idea

3.1 Pipeline Overview

The system is composed of three models, summarized as $A + B \rightarrow C$:

Model	Role
A	Facial expression $\rightarrow$ Emotion (7 classes)
B	Genuine vs. deceptive classification
C	Fusion: true emotion given deception signal

MediaPipe Face Mesh extracts 468 landmark coordinates per frame. These coordinates (not raw pixels) serve as inputs to Models A and B , making both models extremely lightweight and quantization-friendly.

3.2 Semi-Supervised Learning with FixMatch

A key challenge is the limited availability of labeled deception data. We address this with the following four-stage pipeline:

1. **Contrastive pre-training (SimCLR).** Train a feature extractor on *both* labeled (Dataset B : self-collected) and unlabeled data (Dataset A : MU3D used without labels) via contrastive loss. This produces a shared facial-feature embedding space.

2. **Supervised fine-tuning.** Attach a classifier head and train only on Dataset B (labeled), using the frozen SimCLR backbone.
3. **Pseudo-labeling (FixMatch).** Apply the fine-tuned model to Dataset A . Retain pseudo-labels only where prediction confidence exceeds threshold $\tau = 0.95$.
4. **Joint re-training.** Combine pseudo-labeled A with labeled B and retrain the full Model B end-to-end.

This allows us to exploit the large unlabeled MU3D corpus to improve deception detection without requiring manual annotation.

4 Implementation Plan

4.1 Datasets

- **AffectNet:** 450k images, 7 emotion classes $\rightarrow$ Model A
- **MU3D:** genuine/deceptive expression videos $\rightarrow$ unlabeled pool for FixMatch
- **Self-collected:** minimum 20 subjects; genuine vs. posed smiles (5 s clips), controlled lighting/distance $\rightarrow$ labeled set for Model B

4.2 Model Architecture

All three models use **MobileNetV2** backbones fine-tuned on landmark coordinate sequences. Model C is a lightweight MLP that takes the concatenated output logits of A and B as input.

4.3 Model Optimization

Models will be converted to TFLite with **INT8 post-training quantization**. Because inputs are low-dimensional landmark coordinates rather than images, accuracy degradation is expected to be $< 2\%$ while achieving $\sim 75\%$ size reduction — maximizing the scoring criteria for size reduction rate vs. accuracy drop.

4.4 Android App

- Real-time camera feed with MediaPipe Face Mesh overlay
- Live emotion gauge (surface emotion vs. inferred true emotion)
- Deception probability bar; red border alert on high probability
- Post-session summary heatmap of expression timeline

5 Project Timeline

Period	Task
~May 4	Proposal submission; environment setup; dataset download
May 5–11	SimCLR pre-training; Model A & B initial training
May 12–18	Self-data collection; FixMatch pseudo-labeling; Model C fusion
May 19–25	TFLite INT8 quantization; accuracy/size benchmarking
May 26–Jun 1	Android app + visualization; GitHub cleanup; final APK
Jun 2–4	Final presentation

Team Role Division

Member A	Member B
Model training (Python)	Android app development
SimCLR + FixMatch	MediaPipe integration
TFLite quantization	UI & visualization
Self-data collection (joint)	

References

- [1] W. Yan et al., “CASME II: An Improved Spontaneous Micro-Expression Database,” *PLOS ONE*, 2014.
- [2] A. K. Davison et al., “SAMM: A Spontaneous Micro-Facial Movement Dataset,” *IEEE T-AFFC*, 2018.
- [3] E. P. Lloyd et al., “The MU3D: A Multi-Emotion, Multi-Intensity, 3D Face Database,” *PLOS ONE*, 2019.
- [4] K. Sohn et al., “FixMatch: Simplifying Semi-Supervised Learning with Consistency and Confidence,” *NeurIPS*, 2020.
- [5] T. Chen et al., “A Simple Framework for Contrastive Learning of Visual Representations,” *ICML*, 2020.
- [6] C. Lugaresi et al., “MediaPipe: A Framework for Building Perception Pipelines,” *arXiv:1906.08172*, 2019.